

Predictive Power vs. Interpretability

BOTTOM LINE

No — added model complexity did not pay off. Logistic Regression achieved the best result on every core metric, and its ROC-AUC advantage over Naive Bayes, Tree-Augmented Naive Bayes, and a freely-learned General Bayesian Network was statistically confirmed. But the Bayesian approach was not wasted effort: its Markov Blanket analysis independently rediscovered the 5 variables most strongly linked to churn, cutting the feature set by 74% with virtually no loss in predictive performance.

CUSTOMERS ANALYZED

7,043

26.5% churn rate

BEST MODEL ROC-AUC

0.845

Logistic Regression

STATISTICAL CONFIDENCE

3 of 3

comparisons: H0 rejected

FEATURE REDUCTION

73.7%

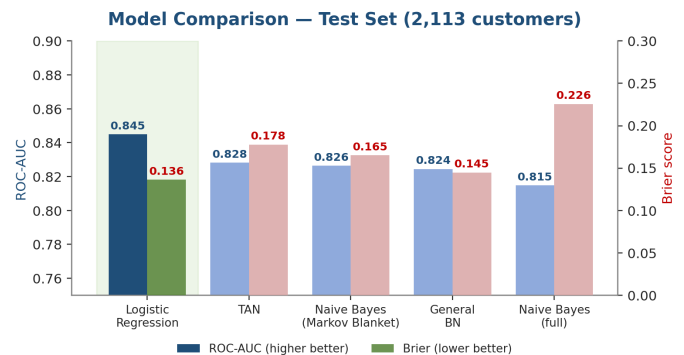
19→5, via Markov Blanket

The Business Question

Bayesian Network classifiers promise something Logistic Regression can't: a visual map of how customer attributes relate to each other, not just to churn. That map is appealing to stakeholders — but only worth the added complexity if it also predicts better. This project tests that trade-off directly, escalating from a simple linear baseline through three Bayesian Network variants of increasing structural complexity.

The answer matters commercially: a simpler, cheaper-to-maintain model that performs just as well should always be preferred — unless the complex model earns its keep with a real performance gain.

Model Comparison



What's in the Full Technical Package

Excel Workbook

Live-formula calculations — confusion-matrix metrics, a TotalCharges data-quality fix, significance verdicts, and dimensionality-reduction figures all recompute automatically.

PDF Dashboard

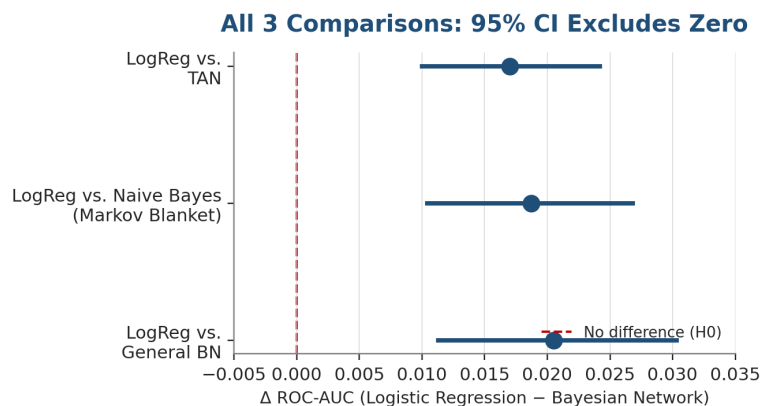
3-page visual deep-dive: model comparison, statistical forest plot, Markov Blanket reduction, and churn-driver breakdowns.

Source Notebook

Full pipeline in Python (pgmpy, scikit-learn) — structure learning, CPT estimation, and paired bootstrap testing, reproducible end to end.

What We Found — and How We Know It Holds Up

Is the Gap Real? Pre-Registered Statistical Test



Before looking at results, we pre-registered a clear test: H_0 = no significant ROC-AUC difference between Logistic Regression and each Bayesian Network variant. A paired bootstrap test (1,000 resamples on the same 2,113 held-out customers) checked each comparison.

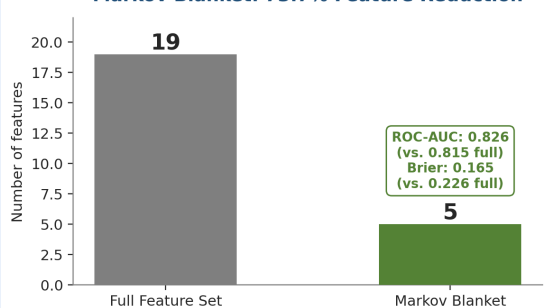
Comparison	Δ ROC-AUC	95% CI
vs. TAN	+0.017	[0.010, 0.024]
vs. Naive Bayes (MB)	+0.019	[0.010, 0.027]
vs. General BN	+0.021	[0.011, 0.030]

All three 95% confidence intervals exclude zero — Logistic Regression's edge is small but statistically real, not a fluke of this particular test split.

Why These Numbers Can Be Trusted

- H_0 pre-registered before results were seen, not fitted to match the outcome afterward.
- Same 2,113-customer test set scored exactly once for all 5 models — a fair, single-look comparison.
- Class-balancing was tested and rejected: it barely moved ranking but meaningfully hurt calibration, so all models use natural class distributions — predicted probabilities are genuine, not artificially reweighted.
- Calibration was checked, not just discrimination: a reliability diagram confirmed Naive Bayes' poor Brier score reflects real systematic overconfidence, not a fluke metric.

Markov Blanket: 73.7% Feature Reduction



The One Place Complexity Paid Off

The General Bayesian Network's freely-learned structure reduced the churn-relevant variable set from 19 to just 5 — Contract, InternetService, PaperlessBilling, TechSupport, and tenure — with no human guidance. Naive Bayes retrained on just those 5 variables matched (and slightly beat) its 19-variable version. 'gender' was excluded from the structure entirely, matching common intuition.

How to Use This — and Where to Be Careful

Practical Applications

Deploy the simple model

Logistic Regression is the production recommendation: equal or better predictive performance, dramatically easier to maintain, retrain, and explain to non-technical stakeholders than a 38-edge Bayesian Network.

Reuse Markov Blanket as a feature-selection tool

Even though the Bayesian Network itself won't be deployed, running a quick Markov Blanket analysis ahead of building future churn models is a cheap, model-agnostic way to validate which features actually matter.

Target retention spend using the 5 confirmed variables

Contract type, internet service, paperless billing, tech support, and tenure — independently confirmed by two different methods as most strongly linked to churn — are where retention budget will have the most leverage.

Where to Be Careful — Limitations

- HillClimbSearch finds a local optimum: a different random seed or scoring method (K2 instead of BIC) could surface a different network structure with different performance.
- Numeric features were quantile-binned for the Bayesian models but left continuous for Logistic Regression — a natural format difference for each model family, not a perfectly neutral comparison.
- Network edges reflect statistical dependency learned from data, not confirmed causal relationships — the 5 Markov Blanket variables are validated predictors, not proven causes of churn.
- Findings are specific to this dataset and a fixed random seed; a dataset with more deeply interdependent features could favor Bayesian Networks more than this one did.

Recommended Next Step

Deploy Logistic Regression to production, using the 5 Markov Blanket variables as the core input for a retention-targeting dashboard. Revisit Bayesian Network structure learning only if a future dataset shows stronger feature interdependence — the added complexity is not justified here.

Part of a Rigor-Driven Analytics Portfolio

This is the fifth project applying Verbraken's (2013) analytics framework to real Kaggle datasets — alongside Telecom Churn Retention (EMP framework), an E-commerce A/B Test (SRM, CUPED), Credit Risk Modeling (leakage correction), and Social Network-Based Classification (homophily, leakage-safe network features). Each project pairs a working model with an equally careful audit of its own assumptions — this one's audit finding is that added model complexity, while intuitive, did not survive a pre-registered statistical test.